A SURVEY ON THE APPLICATIONS OF STOCHASTIC DUAL DYNAMIC PROGRAMMING AND ITS VARIANTS

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ABSTRACT

Stochastic Dual Dynamic Programming (SDDP) is widely recognized as the predominant methodology for solving large-scale multistage stochastic linear programming (MSLP) problems. This paper aims to contribute to the extant literature by conducting a comprehensive survey of the literature on SDDP within the realm of practical applications. We systematically identify and analyze the various domains where SDDP has been successfully employed to tackle MSP problems, with a particular focus on real-world problems afflicted by the so-called *curse-of-dimensionality*. Furthermore, we investigate the factors that have facilitated or hindered the adoption of SDDP in specific application areas, shedding light on the limitations and potential barriers to its widespread utilization.

***K*eywords** Stochastic Dual Dynamic Programming; Approximate Dynamic Programming; Stochastic Programming

1 Introduction

Multi-stage stochastic programming (MSP) is a class of mathematical problems dealing with optimization of sequential decision-making under uncertainty. Solving real-world problems through MSPs presents considerable challenges, particularly when navigating extensive planning horizons and uncertainty scenarios. Such problems frequently run into the infamous *curse-of-dimensionality*, a phenomenon describing the exponential growth in computational requirements as the dimensionality of a certain problem increases.

Enter Stochastic Dual Dynamic Programming (SDDP). Introduced in the seminal work by Pereira and Pinto (1991), SDDP proposes a powerful solution to bypass many of the obstacles associated with MSP problems, particularly the curse-of-dimensionality. Rooted in the principles of dynamic programming and duality, SDDP decomposes an MSP problem, iteratively approximating the expected cost-to-go functions without needing to tackle the entire problem space at once. Consequently, what previously seemed intractable becomes more computationally feasible, allowing practitioners to address large-scale MSP problems in various sectors, from energy planning to supply chain optimization.

The conception of SDDP marked a paradigm shift in the approach to sequential decision-making under uncertainty. Previously, dealing with uncertainty often required, on one hand, simplification and even relaxation of the problem. This potentially leads to suboptimal solutions or ones that failed to mirror the full complexity of a real-world problem. On the other hand, solving the entire scale of MSPs through a tedious exploration and exploitation of the full scenario tree results to expensive computational efforts. However, with SDDP, it became possible to retain much of the intricacies of MSPs while still navigating through the expanse of uncertainties and stages.

Nevertheless, while SDDP is a powerful methodology, it still comes with its challenges and limitations. Affected by factors ranging from computational restraints to domain-specific nuances, SDDP still has a long way to solve even larger problem instances mirroring real-world cases. Füllner and Rebennack (2021) present a more in-depth tutorial-type review of the various improvements and enhancements of SDDP.

To appreciate the relevance of SDDP even further, it is imperative to consider the current challenges faced by different industries worldwide. With increasing globalization and competitiveness, where effects of decisions made today can cascade and amplify over time, industrial applications highlight the invaluable utility of SDDP. For instance, consider the domain of hydro-thermal power system planning, the premier application of SDDP. Governments and private companies across the world harnessing this renewable energy leverage SDDP to aid in their decision-making processes (Maceiral et al. 2018).

Furthermore, as the eminence of SDDP continues to advance, its applications have prospered across a wide array of applications. Nonetheless, a significant proportion of the work in the extant literature is primarily focused within the field of energy planning. This survey endeavors to explore the breadth and depth of the applications underlining the significance of SDDP and its growing relevance in solving MSP problems.

Through this review, we aim achieve two goals. First, to provide a comprehensive survey of the practical applications of SDDP in the extant literature. We deep-dive into this exploration of the literature in the lens of these different applications. Second, we aim to present a compelling discussion on the factors that have facilitated and, possibly, hindered the adoption of SDDP in specific application areas shedding light on the limitations and potential barriers to its widespread utilization. As we delve deeper into the sphere of SDDP and its applications, it is significant to understand not just where and how SDDP shines, but also where it can falter and require supplemental schemes. This perspective is crucial for both practitioners and researchers aiming to augment the boundaries of solving MSP problems.

The remainder of the paper is organized as follows. Section 2 provides an overview of the history, evolution, and technicalities of SDDP. Section 3 covers the literature selection process. Section 4 constitutes one of the contributions of the paper and discusses the main statistical findings from our literature survey. Section 5 discusses various factors of adoption of SDDP in specific application areas. Section 6 contains some concluding remarks and provides guidance as to future research directions.

2 Background

Before the inception of SDDP thirty years ago, the method draws upon a rich lineage of inspirations and developmental precursors. In this section, we will briefly revisit three pivotal stepping-stone methods that laid the foundation for the evolution of SDDP: (a) Benders decomposition, (b) L-shaped method, and (c) nested Benders decomposition. We will then proceed to formally introduce SDDP and its variants and extensions.

2.1 Overview of the History and Evolution of SDDP

The origins of SDDP can be traced back from the groundbreaking method developed by Jacques Benders (Benders 1962) - Benders decomposition method. Benders decomposition method is a decomposition method for solving mixed integer programming problems - optimization problems where some variables are constrained to take real values and others only limited to integer values. The basic idea is to decompose the problem into two: a pure linear programming problem [called master problem (MP)] and a pure integer programming [called subproblem (SP)]. Benders decomposition proceeds by solving MP and SP iteratively to arrive to a solution for the overall problem by adding $Benderscuts$, some feasibility or optimality information from the sub-problem to the master problem. The leverage of the algorithm is the fact that two separate decomposed problems are more manageable to solve than the original mixed problem. Notwithstanding the number of iterations to solve each subproblem, this technique is still relatively quicker to arrive at the solution than solving the large mixed integer programs.

Seven years later, Van Slyke and Wets (1969) expanded on this powerful algorithm and introduced the L-Shaped method. L-shaped method is an algorithm to solve stochastic programming problems – optimization problems where some, if not all, of the problem parameters take stochastic or uncertain values. For L-Shaped method, two-stage stochastic problems or stochastic linear problems with recourse are of interest where an initial decision is made succeeded by a second decision which allows remedial (recourse) action after an event have been observed to happen. Similar to Benders' decomposition, L-Shaped method is based on adding additional constraints (or cuts) to the original (in this case, first-stage) problem. These cuts ensure feasibility of the first-stage decision to the second-stage problem and assurance to obtain an optimal solution.

Finally, the nested Benders decomposition (NBD) algorithm from Birge (1985) is an extension to L-Shaped decomposition such that the former proposes an algorithm for multi-stage stochastic programming problems. The simple idea behind nested Benders' decomposition is the notion of decomposing multi-stage problems into nested two-stage problems and recursively, then solving these with L-shaped method over the entire scenario tree structure. With the initial decision of the first-stage (root node), succeeding second-stage sub-problems will be solved. Thereupon, the initial decision will be viewed as a constant and the following stage nodes will become the first-stage (root node) of its own problem. This will continue further until the end of the planning horizon. As with L-shaped method, NBD solve the sub-problems with the current initial decision and use these solutions to create feasibility and optimality cuts. This means that parts of the problem of whichever stage of the tree can be solved as long as initial decision of a preceding node exists. The drawback of NBD is that the scenario tree grows exponentially in the number of stages and it becomes computationally expensive for large instances.

2.2 SDDP Methodology and some extensions

SDDP is a straightforward methodology for solving MSP problems by approximating the expected cost-to-go value functions of stochastic dynamic programming using piece-wise linear functions. Similar to Benders decomposition algorithm, SDDP algorithm incorporates *Benders cuts* – which are approximations of the second stage (or the next consecutive stages in the case of a multi-stage problem) – into the first stage (precedent or root) problem. These cuts provide upper and lower bounds for the optimal solution and are derived from the dual solutions of the problem of the succeeding stages.

As mentioned, one of the significant advantages of SDDP is its avoidance of the curse-of-dimensionality associated with dynamic programming. With a few iterations, the upper and lower bounds converge, providing an optimal solution (or a near-optimal solution, based on a given tolerance). This convergence makes SDDP an efficient and effective approach for solving MSPs (Philpott and Guan (2008), Girardeau et al. (2015), and Guigues (2016)). The idea of SDDP stems from NBD but it includes the concept of sampling. In each iteration, not all the scenarios of the stochastic data process scenario tree will be explored but only a sample of those scenarios will be drawn and used. This will significantly reduce the number of linear programs to be solved.

The vanilla SDDP algorithm rests upon several assumptions for its effective implementation. However, numerous refinements and advancements have emerged in recent years, enhancing the original method. As an illustration, an extension to the vanilla SDDP is the incorporation of stage-wise dependent uncertainty of the MSP. The original SDDP limits the problem in consideration to have its random data process stage-wise independent - the case where the uncertainty in different stages do not depend on each other. In this case, the respective value functions of the dynamic programming equations are independent of the data process. Löhndorf and Shapiro (2019) introduce two approaches: (1) the modelling of the random data process as an *auto-regressive time series* by adding state variables to the model and (2) the use of Markov chain discretization of the random data process. Additionally, the pioneering works of Guigues and Römisch (2012b), Guigues and Römisch (2012a), and Shapiro et al. (2013a) stand out as the initial trio of papers that expanded upon the original assumption of risk neutrality in MSP problems for SDDP. Their contributions extend towards encompassing various risk-averse problems.

Furthermore, SDDP has been extended to tackle multistage stochastic integer programs (MSIPs). Stochastic Dual Dynamic Integer Programming (SDDiP) stands out as the primary computationally tractable method to obtain both a statistical lower bound and an optimal policy (Zou et al. 2019). As with SDDP, SDDiP involves a reformulation of the MSIP into a dynamic program. Under specific assumptions, SDDiP guarantees convergence to an optimal policy. Like SDDP, SDDiP assumes that the stochastic parameters in the MSIP follow a discrete distribution. A key ingredient to SDDiP is the binarization of all time-coupling state variables. By incorporating binary variables, SDDiP ensures the efficient representation and exploration of the decision space, resulting in a more manageable problem formulation.

Finally, the evolution of SDDP remains an ongoing development, marked by continuous refinement and enhancement. Recent efforts are focused on finding of optimal policies quickly and extending its applicability to a wider array of real-world problems.

3 Literature Selection and Analysis

We conducted a comprehensive database search for papers that apply, utilize, develop, or enhance SDDP to primarily address optimization problems. The databases examined included Emerald Insight, Scopus, JSTOR, and Web of Science which included published journal articles until December 2022. We employed the keywords "stochastic dual dynamic programming," "SDDP," "stochastic dual dynamic integer programming," and "SDDiP." Publications not written in English, as well as those exclusive to conference proceedings and book chapters, were excluded from our consideration.

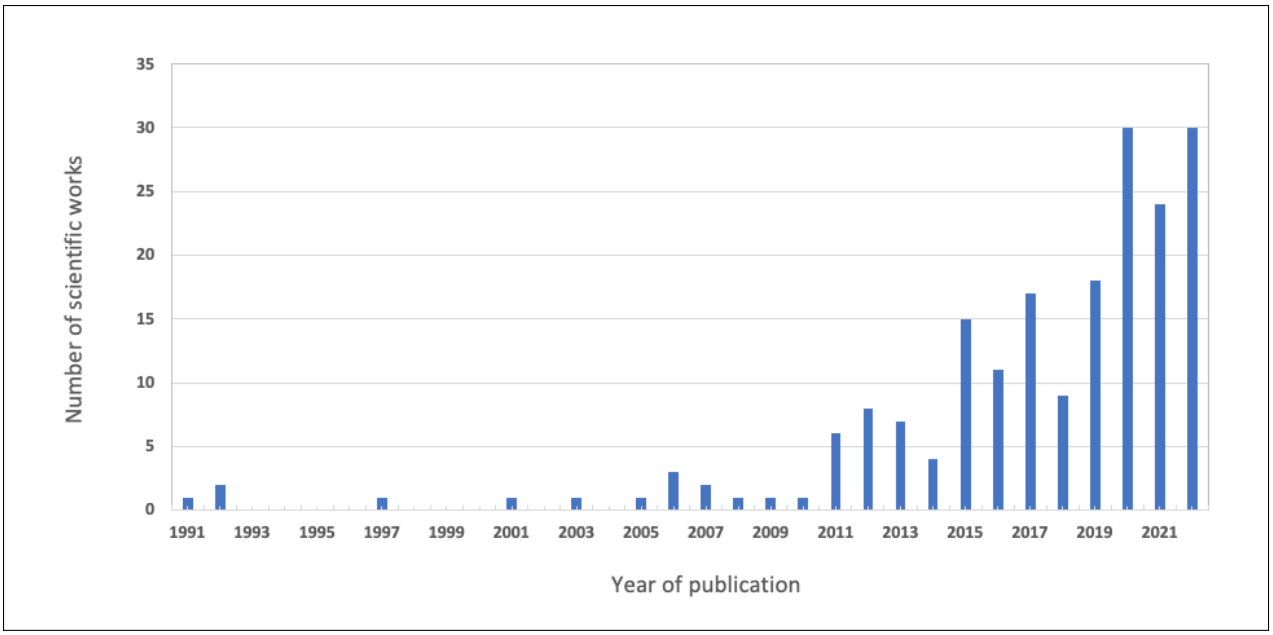


Figure 1: Evolution of number of published works over time.

Our approach to filtering relevant publications began with an evaluation of the title. If deemed pertinent, we proceeded to assess the abstract, followed by the introduction and conclusion. Only if the paper still appeared relevant after this thorough review did we analyze its entirety. This meticulous method led to the identification of 186 significant publications since the release of the seminal paper on the subject in 1991. The collective works surveyed are graphically illustrated in Fig. 1. These papers span 61 distinct journals, which are itemized in Table 1, covering a spectrum of research domains. We further dissected the significance of these scientific contributions by examining three primary research questions:

1. In which specific optimization problems is SDDP employed, and within which application sectors (e.g., energy, finance, operations management)?

2. What are the key contributions of the research to SDDP (be it theoretical, methodological, or other)?

3. How valuable is SDDP in efficiently resolving the identified problem?

Our survey unveiled several distinct fields of application. Predominantly, a significant portion of the papers concentrates on issues within the energy sector. The Brazilian hydro-thermal power system is a primary focus in this area. This problem manifests in various iterations — different planning horizons, diverse geographical locations, and a range of different random data processes, to name a few. A secondary line of literature navigates challenges in the financial sector, such as portfolio optimization and asset allocation. Operations management is another prominent domain addressed, with a unique set of scientific works exploring issues like inventory, lot sizing, and production. Additionally, certain studies venture into more specialized areas, delving into matters related to disaster management and mining.

4 Literature Review and Findings

In this section, we provide an extensive analysis of various applications of SDDP as found in the extant literature. This examination is structured into distinct subsections that correspond to different fields of application. Notably, the field of energy applications, with its broad scope, is itself subdivided into various domains for a more comprehensive discussion. The distribution of works across these application areas is illustrated in Figure 2. While some studies are inherently dedicated to a specific application field, a significant number of papers span multiple domains. For instance, a work might address an energy system application while also incorporating important financial aspects. In such instances, we ensure that these papers are discussed and integrated into both [or more] relevant subsections.

Table 1: Classification of the papers according to the different journals.

Journal	Number
European Journal of Operational Research	19
IEEE Transactions on Power Systems	14
Mathematical Programming	10
IEEE Transactions on Sustainable Energy	9
Journal of Water Resources Planning and Management - ASCE	9
International Journal of Electric Power and Energy Systems	9
SIAM Journal on Optimization	9
Electric Power Systems Research	7
Water Resources Research	7
Computational Management Science	7
INFORMS Journal on Computing	5
Annals of Operations Research	5
Computational Optimization and Applications	5
Operations Research Letters	5
Optimization and Engineering	4
Energy Policy	4
Operations Research	4
Energies	4
IEEE Transactions on Smart Grid	4
IET Generation, Transmission, and Distribution	3
Others	43

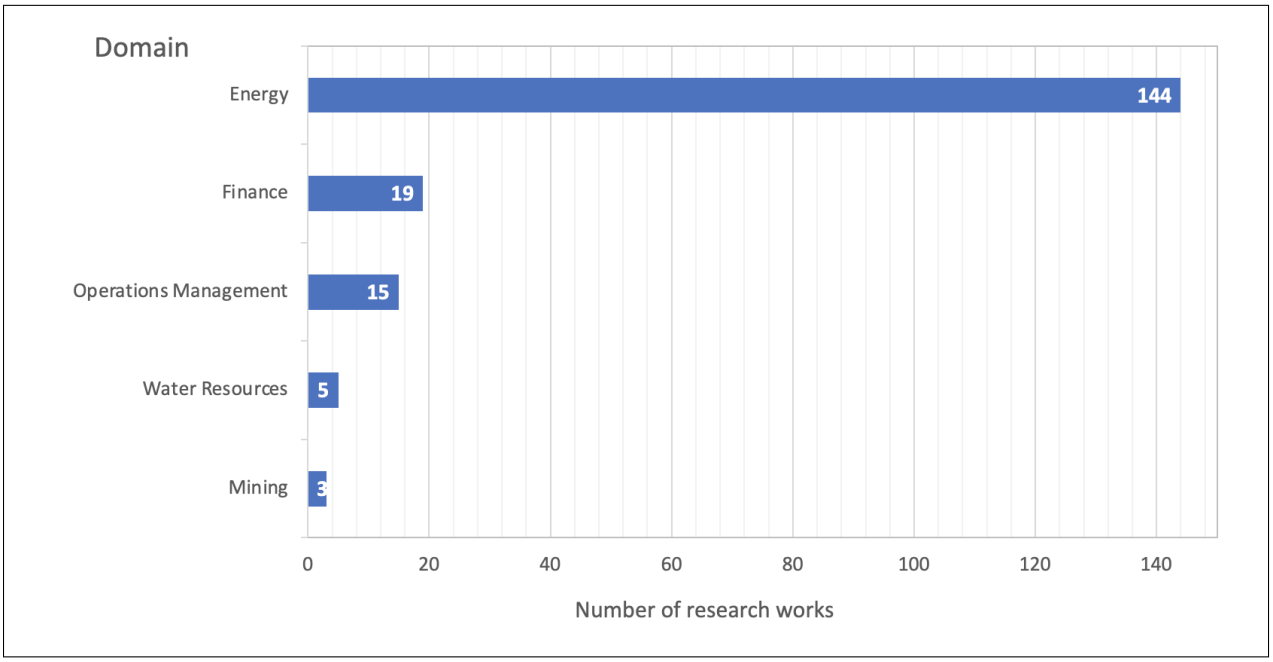


Figure 2: Distribution of research works across different application areas.

4.1 Energy

With reference to Figure 2, it is evident that the energy sector constitutes a majority of the research works focused on SDDP. This section explores diverse domains within the energy sector - applications related to hydro-thermal power system, problems associated with short-term economic dispatch and capacity expansion, the integration of various renewable energy sources, and optimization problems arising in energy markets, trading, and investments. Additionally, the integration of different energy storage systems and the intricacies of grid management are also discussed thoroughly.

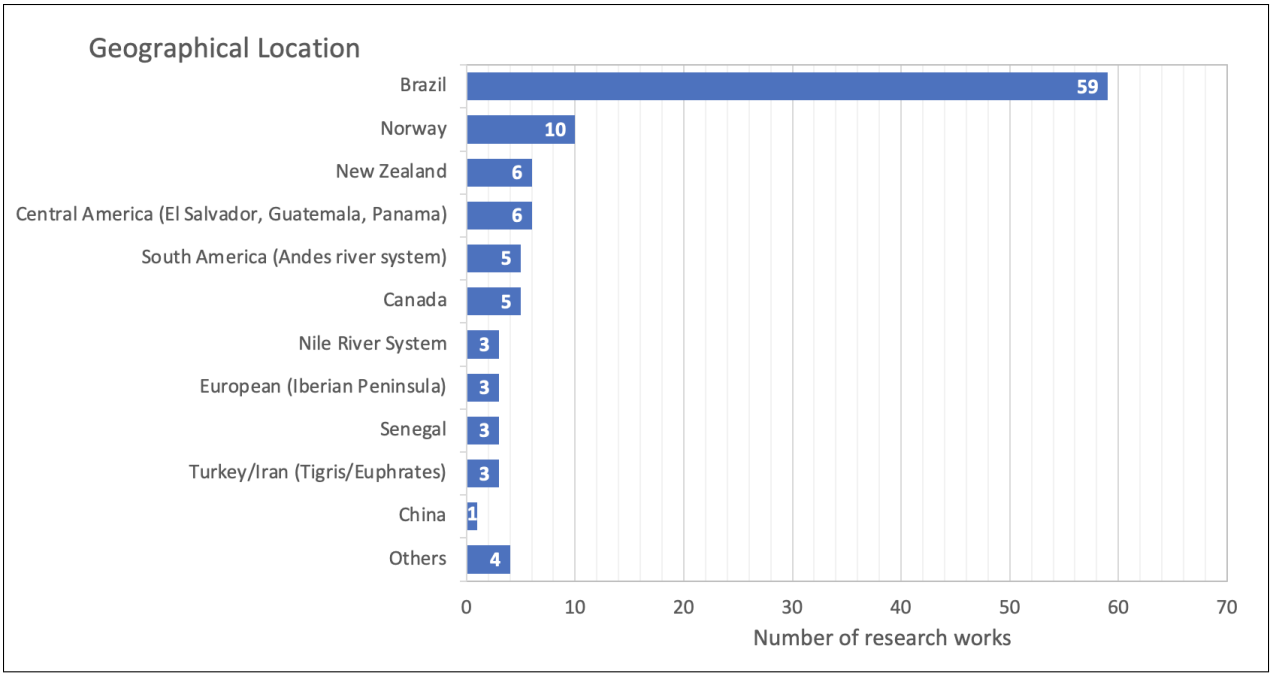


Figure 3: Distribution of research works across different geographical locations for hydro-thermal power production problem.

4.1.1 Hydro-thermal Power Production Problem

The hydro-thermal power production serves as a flagship problem application of the development of SDDP. The central question in this domain revolves around the optimal coordination between hydroelectric and thermal power generation sources to meet demand while minimizing costs and maintaining reservoir storage, considering the unpredictable nature of water inflows into hydroelectric reservoirs, and the fluctuating costs of thermal power generation. Managing these resources is crucial not just for economic efficiency but also for maintaining overall system reliability. Given the complexity arising from the multitude of reservoirs, interconnected systems, and uncertainty of inflows, traditional deterministic methods often fall short in providing practical solutions. This is where SDDP becomes invaluable.

Since the introduction of SDDP over 30 years ago, with the Brazilian interconnected hydro-thermal power system serving as its premier case study, over 100 research publications have been dedicated to addressing the hydro-thermal power production problem using this method. The variations encountered within this domain typically stem from differences in planning horizons, modifications in the uncertainty processes, and diverse geographical contexts. Figure 3 presents the distribution according to various geographical locations, while Figure 4 illustrates the distribution of research works based on distinct planning horizons. A detailed breakdown, encompassing planning horizons, geographical areas, and uncertainty processes related to the hydro-thermal power production problem, can be found in Tables B and B in the Appendix. For an in-depth discussion in this section, we categorize the literature based on temporal scales, which offer more consistent comparisons due to the similar objectives of the problems addressed. The distinct operational planning horizons are as follows: (1) long-term operational planning, covering a span of 3 or more years with monthly decision epochs; (2) medium-term operational planning, typically lasting 1 to 2 years with weekly or monthly decision epochs; and (3) short-term operational planning, extending over hours or days.

Long-term Operational Planning. Long-term operation planning primarily focuses on multi-annual horizons (3 or more years), usually with a monthly decision epoch, ensuring that the system is resilient against seasonal variations and long-term climate changes. The main objective of the solving this problem is to ensure strategic reservoir storage decisions, ensuring that water resources are adequately maintained to meet future demands.

Furthermore, with the seasonal uncertainty of water inflows, long-term planning ensures that reservoirs maintain optimal levels especially in handling drought and surplus events. This helps in preventing potential water shortages in dry periods and avoids overflow during wet periods. This is the reason that all, if not majority, of research works are devoted to understanding the inflow uncertainty. Sampling from the historical values is one of the easy go-to methods in choosing realizations of inflow uncertainty. Nonetheless, with the understanding that inflow has seasonal patterns,

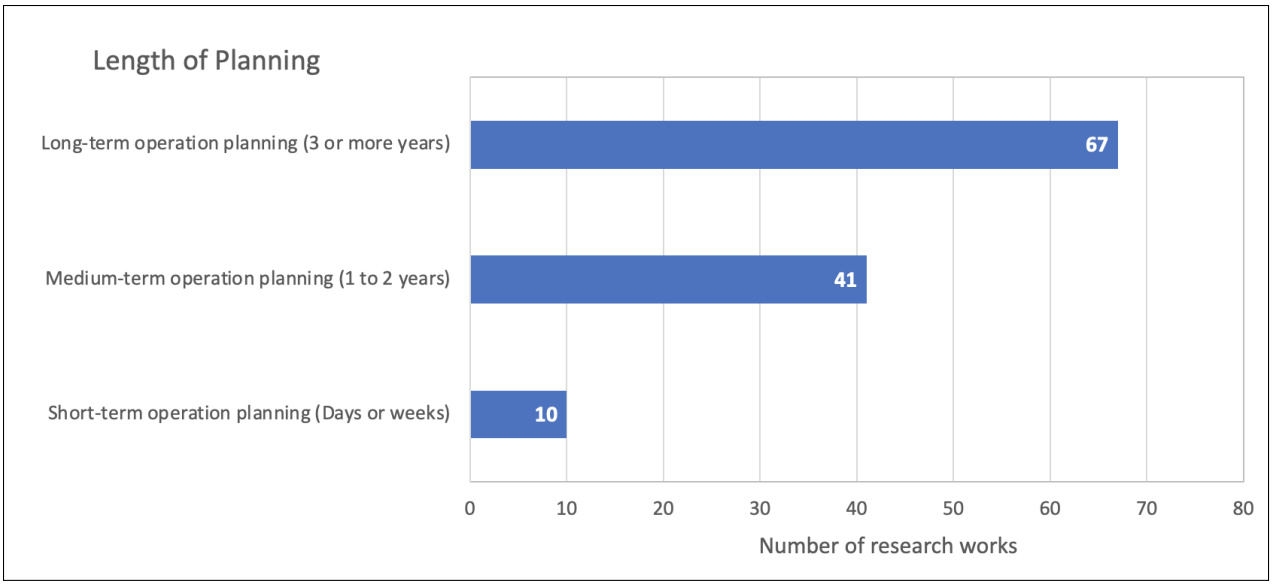


Figure 4: Distribution of research works across different planning horizons for hydro-thermal power production problem.

forecasting techniques were also adapted to understand this random process. Various aut-oregressive processes and the inclusion of non-linearities are among the prominent techniques employed in this end.

Long-term planning also involves prediction of future energy demand. Some factors that affect demand include population growth, industrial and local development, and economic trends. Forecasting demand correctly aids in ensuring the optimal operation of the system without frequent and costly re-adjustments. For example, works in Shapiro and Cheng (2021), Raby et al. (2009), and Sauma et al. (2011) include demand uncertainty in their models.

A phenomenon in the long-term operation planning of hydro-thermal power production worth mentioning is the *end-of-horizon* effect. Policies derived from optimization models do not necessarily ensure a sustainable and reliable energy supply beyond the designated planning horizon. The reason being, optimal policies typically exhaust all energy reserves in the reservoirs at the end of the planning horizon. Thus, most works consider doubling the planning horizon (using 10 years instead of 5 years) in order to hedge from this phenomenon.

As mentioned, a more detailed breakdown, encompassing planning horizons, geographical areas, and uncertainty processes related to the hydro-thermal power production problem, can be found in Table in the Appendix.

Medium-term Operational Planning Medium-term operation planning serves as a link between the expansive outlook of long-term operational planning and the immediate requirements of short-term operational planning. Typically encompassing a 1 to 2-year planning horizon, medium-term operation planning operates with a monthly or weekly decision epoch, allowing for a more granular view of the system's operations than long-term planning.

As hydro-thermal power systems cope with the unpredictability of inflows and other uncertainties, medium-term operation planning offers flexibility in accommodating issues that long-term operation planning usually overlooks. One example is the incorporation of planned maintenance schedules to ensure that energy production and distribution assets are kept at optimal levels. Additionally, medium-term operation planning is significant in response to unexpected disturbances in the system like technical malfunctions, sudden demand surges, or other external factors.

Furthermore, works for medium-term operation planning often include the uncertainty of prices and costs. This provides producers, not only to be economically profitable with their operations, but also to perform optimally in energy trading. For example, the works in Kristiansen (2006), Löhndorf et al. (2013), de Matos et al. (2015), Helseth et al. (2016), Rebennack (2016), Hjelmeland et al. (2018), Hjelmeland et al. (2019), and Borges et al. (2022) include uncertainty in prices and costs for the medium-term operation planning of hydro-thermal power system.

Water Resource Management Given the pivotal role of water resources in hydro-thermal power systems, making strategic reservoir storage decisions is crucial - ensuring that water reservoirs are optimally maintained, without compromising the ability to meet future energy demands. Hence, in the operational planning for hydro-thermal power production, the effective and sustainable management of these water resources is important.

Various research works have been directed towards understanding and optimizing water resource management, particularly in the context of major river basins integral to hydro-thermal power systems. Notably, African river basins have been the subject of study, with Rougé and Tilmant (2016) looking at the Zambezi river basin, while Raso et al. (2017) focus on Mananatili river basin. In Europe, the Iberian peninsula, a significant hydro-thermal hub, is the subject of study in Pereira-Cardenal et al. (2016) and Macian-Sorribes et al. (2017). Shifting to the Middle East, Rougé et al. (2018) and Tilmant and Kelman (2007) explore the Tigris-Euphrates river basin considering its pivotal role in the region's energy landscape. Moreover, Treistman et al. (2020) propose innovative synthetic inflow scenario generation techniques tailored to the El Niño–Southern Oscillation, offering insights for the Brazilian hydro-thermal power system.

Short-term Operational Planning - Economic Dispatch and Capacity Expansion The smallest coverage of time horizon is the short-term operation planning which zeroes in on daily or hourly scales. It focuses on fine-tuning generation schedules to respond to daily demand variations. Two domains are included in this area - economic dispatching and capacity expansion planning. The difference between the two lies in the objectives of the operation and planning of any power systems. Economic dispatch focuses on the short-term optimization of existing power generation units to meet the current electricity demand at minimum cost. It aims to determine the optimal output levels of each generator to match demands, while considering different factors such as fuel costs, transmission constraints, and generator constraints. On the other hand, capacity expansion planning deals with determining optimal additions or retirements of power generation units to meet future demand.

The following works have made notable contributions to various aspects of short-term economic dispatch problems: Chabar et al. (2006), Papavasiliou et al. (2017), Stüber and Odersky (2020), Ding et al. (2021), Lan et al. (2022), and Pacaud et al. (2021). Moreover, economic dispatch has been extended to incorporate additional system dynamics. For instance, Lu et al. (2019) integrate energy reserves, while Street et al. (2017) delve into ancillary services. Additionally, Fatouros et al. (2017) explore the operation of distributed energy resources.

Turning to generation expansion problems, researches and works encompass diverse perspectives. da Costa et al. (2020) include reliability constraints using a stochastic risk-averse approach, Ihsan et al. (2021) assess the performance of distributed hybrid renewable power plants in both on-grid and off-grid contexts, and Rebennack (2014) introduce the incorporation of emissions quotas as a critical factor.

4.1.2 Renewable Energy

With the significant attention that production planning for hydro-thermal power, a prominent renewable energy resource, has gained in SDDP research, it is only natural for similar focus to extend to other renewable energy sources. For instance, a couple of works have broadened the scope of hydro-thermal power production problem to incorporate additional renewable energy sources. As initiated in Raby et al. (2009) wind farms have also been integrated into the hydro-thermal power transmission system. Likewise, various other studies such as de Faria and Jaramillo (2017), Alvarez et al. (2018), da Silva Fernandes et al. (2019), Scheben et al. (2020), and Morillo et al. (2020) explore the coupling of hydro-thermal power and wind energy. Moreover, the integration of wind energy production has also been linked with power grid management and storage in Zéphyr and Anderson (2018), and with microgrid systems with storage in Guo et al. (2020). Wozabal and Rameseder (2020) address uncertainty in wind energy production for optimal bidding in the Spanish electricity market.

Other applications integrating renewable energy sources include the economic dispatch problem with different renewable energy resource uncertainties as in Lu et al. (2019) and in Papavasiliou et al. (2017) for storage. Bao et al. (2019) investigate the optimization of renewable bioresources, and cogeneration biomass plants in Leocadio et al. (2020). Finally, Pacaud et al. (2022) and Aaslid et al. (2022b) discuss a complex combination of renewable energies and storage to optimize microgrids.

4.1.3 Energy Markets, Trading, and Investments

In addition to optimizing aspects such as planning, dispatch, and the expansion of energy systems, another critical domain of optimization involves the financial facets of energy – bidding, trading in energy markets, and investments. Energy markets present challenges marked by fluctuating market-clearing prices, dynamic demand patterns, and uncertain renewable energy production. Understanding energy market dynamics provides a framework to optimize bidding strategies capable of robust adaptation to these uncertainties. Trading strategies empower market participants to make real-time decisions that balance profit maximization and risk aversion. Moreover, the use of SDDP provides the ability to make dynamic and well-informed decisions.

The work in Bonnans et al. (2012) initiated this field of application by addressing the optimal management of energy contracts. The model specifically provided bounds on the local and global quantities to be traded, considering uncertainties in prices. This is followed by Bruno et al. (2016) with a development of an MSP model to determine

optimal strategies for energy project investments with real options such as postponing, hedging with fixed forward contracts and other sources.

Covering a range of optimal strategic bidding problems, Wozabal and Rameseder (2020) and Shinde et al. (2022) delve into day-ahead and intraday bidding strategies for electricity within a virtual power plant framework, incorporating renewable energy sources. Steeger and Rebennack (2017) address a single-price maker strategic bidding problem, backed by case studies involving El Salvador, Honduras, and Nicaragua. Shifting focus to the domain of natural gas storage valuation, Löhndorf and Wozabal (2021) investigate this issue through an asset-backed trading approach that takes into account agents' risk preferences.

In the context of emissions trading, Zapletal et al. (2022) introduce a multi-stage planning approach involving carbon risk management and the innovative EU emissions management scheme. Their study compares EUA futures (derivatives) and banking. On a broader scale, Scheben et al. (2020) examine the impact of inflow randomness, revealing that a substantial integration of energy storage devices and renewable energies significantly influences electricity price structures within markets. This observation is supported by a case study conducted in Norway.

4.1.4 Energy storage systems

As the energy sector gravitates towards a more decentralized and renewable-centric paradigm, energy storage devices have emerged as viable solutions in energy systems, particularly in their ability to bridge the gap between intermittent energy production and demand uncertainty. These devices, from advanced battery systems to pumped hydroelectric storage, offer the capability to capture energy produced at a certain stage and utilize it at a later stage. Within the context of energy, MSP models have been instrumental to the optimal management and utilization of these storage devices become predominant.

The integration of energy storage devices into management of energy systems poses new challenges and opportunities, and their exploration in the literature is relatively recent. Distributed energy storage modelled with MSPs were first addressed in Gangammanavar and Sen (2016). Building on this, Zéphyr and Anderson (2018) incorporate wind energy uncertainty considerations for power grids.

Furthermore, within the broader spectrum of grid management leveraging the MSP framework, Asamov and Powell (2018) investigate energy storage in large transmission grids, while Bhattacharya et al. (2016) and Aaslid et al. (2022b) focus on optimizing energy storage in microgrids. Further advancements in microgrid systems is showcased by Pacaud et al. (2022) and Hafiz et al. (2019b), with a particular emphasis on integrating solar panels and batteries. Moreover, Guo et al. (2020) introduce a robust, data-driven approach to optimizing microgrids that harness wind energy, and Aaslid et al. (2022a) consider battery degradation.

Other applications of energy storage devices in energy include work in Ding et al. (2021) illustrating the concept of virtual energy storage using distributionally robust SDDP. On the matter of gas storage, Terça and Wozabal (2021), van Ackooij and Warin (2020), and Löhndorf and Wozabal (2021) examine its pricing and valuation. Lastly, Alvarez et al. (2017) discuss hydroelectric storage in conjunction with batteries, and in Guan et al. (2018) concerning flood control applications.

4.1.5 Grid management

Electrical grid management has also received quite an attention for SDDP application. Works in Asamov and Powell (2018) and Yang and Nagarajan (2022) delve deep into large power grid management. Ding et al. (2020) and Lara et al. (2020) extend this with MSIP formulation solved by SDDiP. Additionally, Zéphyr and Anderson (2018) incorporate wind energy and storage elements, while Hafiz et al. (2019a) discuss the enhancement of grid resiliency against natural calamities.

Meanwhile, the realm of microgrid management, Bhattacharya et al. (2016), Ding et al. (2021), Pacaud et al. (2022), Guo et al. (2020), Aaslid et al. (2022b), and Aaslid et al. (2022a) expansively cover its synergy with energy storage devices. Diving deeper into this area, Shi et al. (2019) explore the integration of varied distributed energy resources within microgrids. Furthermore, Carpentier et al. (2020) present an examination of an urban microgrid problem, while Pacaud et al. (2021) emphasize the aspect of economic dispatching within these systems.

4.2 Finance

Within the realm of finance, the MSP framework seamlessly aligns with the challenges of optimizing portfolio and asset investments. The seminal work presented in Dantzig and Infanger (1993) offers a comprehensive introduction to modeling portfolio optimization, commonly known in the literature as asset allocation, modelled as an MSP. A crucial feature of multi-stage asset allocation models is the investor's stage-wise capability to recalibrate the portfolio,

either by selling or purchasing specific assets, facilitating an ongoing optimization process. The uncertainties of this problem often revolve around the stochastic returns of the chosen investments, e.g., stocks and bonds, and the associated borrowing costs incurred when a specified financial target is elusive. Asset allocation problems, especially when considering large number of assets or over extended time horizons, can exhibit high dimensionality which present a computational challenge for SDDP method.

Furthermore, understanding the behaviour of investors is inherently challenging. Investors naturally gravitate towards risk-aversion to effectively safeguard their investments, which also increases the difficulty of the problem. Homem-de Mello and Pagnoncelli (2016) comprehensively address the formal introduction of risk measures into MSPs, a methodology to which SDDP is easily adapted. Prior to this, three seminal works incorporate risk-aversion into portfolio optimization using SDDP. Specifically, Dupačová and Kozmík (2015) present a risk-averse MSP underpinned by a coherent risk measure. In contrast, both Kozmík (2015) and Kozmík and Morton (2015) apply a risk-averse formulation, but grounded in conditional value-at-risk as the risk metric. Subsequent studies, including Guigues (2017), Valladão et al. (2019), Waga et al. (2022), and Dowson et al. (2022b), built upon these foundational works.

The asset allocation problem has further evolved, embracing diverse contexts. Reus and Prado (2022) delve into large-scale asset allocation, optimizing policies aligned with user-defined (synthetic) indices. Silva et al. (2021) solve a dynamic asset allocation case study underpinned by a data-centric prescriptive analytics framework. Meanwhile, Tsang et al. (2022) put forth a distributionally robust multi-period portfolio model, addressing ambiguity in asset correlations while maintaining fixed metrics for individual asset return mean and variance.

From a technical perspective, asset allocation has also served as a platform to demonstrate advancements in SDDP techniques. For instance, Guigues (2014) present formulas that distribute optimality and feasibility cuts across nodes within the same stage. Various SDDP variants were introduced by Guigues et al. (2020), Guigues (2020), and Dupačová and Kozmík (2017), including regularized, inexact, and preprocessing via scenario tree reduction, respectively. Guigues and Monteiro (2021) add to the discourse with the stochastic dynamic cutting plane (StoDCuP) extension to SDDP. Bandarra and Guigues (2021) outline the nuances of single-cut and multi-cut SDDP, coupled with cut selection for asset allocation. Finally, Guigues (2021) tackle MSPs characterized by an uncertain number of stages.

4.3 Operations Management

Outside of energy and finance, operations management also presents an intriguing application area for SDDP. Domains within this sphere encompass inventory management, production and manufacturing, and facility location, among others.

In the realm of inventory management, variations of the classical problem have been explored in references such as Parpas et al. (2015), Guigues (2017), Dowson et al. (2020), Shapiro and Cheng (2021), Ávila et al. (2022), and Guigues et al. (2023). Lot-sizing problems have also received attention. For instance, Ahmed et al. (2022) address a capacitated version, Quezada et al. (2022a) discuss the uncapacitated variant, and Thevenin et al. (2022) investigate a version incorporating component substitution.

Shifting the focus to production and manufacturing, several studies stand out. Nannicini et al. (2021) tackle bike production with uncertain demand, incorporating aspects of backlog and re-balancing. Yıldız and Sütçü (2022) work on optimal pricing strategies for slow-moving items, while Tong et al. (2020) approach the challenges of central kitchen production and distribution. The pastoral dairy farm planning has also been delved into, with Dowson et al. (2019) and Dowson (2020) addressing specific instances. Moreover, Dowson et al. (2022b) provide insights into a particular routing problem.

4.4 Other areas and works

While energy, finance, and operations management dominate as the primary sectors for SDDP application, emerging innovative problems are gaining attention as new application areas. Notably, disaster management has begun benefiting from SDDP. For instance, Angün (2015) employs SDDP to address short-term disaster management and Hafiz et al. (2019a) harness SDDP to enhance grid resiliency against natural disasters through distribution service restoration. Raso et al. (2019) also apply SDDP for reservoir management—specifically focusing on drought and flood protection in the upper Seine-Aube river system. Furthermore, in the mining industry, Reus et al. (2019) leverage SDDP to devise optimal policies for more effective management of production incidents.

4.4.1 Stochastic Dual Dynamic Integer Programming (SDDiP)

In the context of SDDiP, several studies stand out for their noteworthy contributions. For instance, both Zou et al. (2019) and Ding et al. (2019) delve into issues surrounding airline revenue management. The classic OR problem of facility

location has also seen innovative applications in this domain. Yu and Shen (2022) explore a variant of the capacitated facility location problem, introducing elements of demand uncertainty. Meanwhile, Seranilla and Löhndorf (2023) not only implemented SDDiP for a COVID-19 vaccine facility location instance but also contrast it with a newly proposed methodology.

Additionally, Quezada et al. (2022b) venture into the complexities of remanufacturing under uncertainty. Bakker et al. (2021) solve the problem of optimal timing of investments in mature oil and gas fields in the presence of price uncertainty, like a complex real options problem. For energy planning, Lara et al. (2020) and Hou et al. (2021) present a model on electric power infrastructure planning under uncertainty and Zou et al. (2018) solve a multi-stage stochastic unit commitment problem.

5 Discussion

In this section, we delve into the various factors that influence the adoption of SDDP in certain application domains. Our goal is to highlight the challenges and potential obstacles to its more widespread use. To provide a comprehensive perspective, our conclusions draw from both the extensive literature review in Section 4 and engaging discussions with prominent researchers in the SDDP and stochastic programming field. Specifically, we have pinpointed four key factors that play a pivotal role in the adoption of SDDP: (1) historical background, (2) state space limitation, (3) stage-wise independence assumption limitation, and (4) data availability.

5.1 Historical background

The success of SDDP in addressing its flagship application, the Brazilian hydro-thermal power production problem, in the seminal paper by Pereira and Pinto (1991), played a significant role in its widespread adoption, largely in other energy-related applications. Thus, in the broader academic and research communities, SDDP's evident success on this problem became a compelling narrative. This prompted a surge of literature on the subject. Many researchers were drawn to the challenge of refining the method, expanding its applications, or tailoring it for specific regional or technical nuances. This is highlighted by the literature - from the year of SDDP's inception in 1991 until 2016, all published works in the energy sector that utilized SDDP were centered exclusively on the hydro-thermal power production scheduling problem.

This historical synergy between SDDP and the energy sector has profoundly shaped the trajectory of research and practical applications in this field. SDDP's resilience in handling large-scale, multi-stage stochastic optimization problems, a characteristic that is often inherent in many problems in the energy sector underscores why a bulk of literature on SDDP is anchored in this field of application.

Over time, as the challenges of the energy sector evolved and expanded, the versatility of SDDP kept pace. It became the go-to methodology for many researchers, leading to a wave of academic publications and practical applications. Its historical successes in the energy sector spurred curiosity, experimentation, and eventually, a prolific presence in scholarly discourse - even parallel sessions in conferences are devoted to SDDP in the energy sector.

Currently, when reviewing literature related to optimization, in the context of energy, the echoes of SDDP's historical triumphs are indisputable. Unfortunately, while it testifies to SDDP's efficacy in energy-related applications, it has inadvertently created a perception among some that SDDP is exclusively tailored for energy problems. This misconception limit the exploration and adoption of SDDP in diverse fields, potentially sidelining its benefits in contexts beyond energy.

5.2 State space limitation

At its core, SDDP is sophisticated and manages to solve intricate optimization problems, and it has showcased exemplary efficiency. While SDDP undeniably holds merits in solving many problems, its inherent limitations become more pronounced as the scale and complexity of problems increase. A fundamental assumption of SDDP is that consecutive stages should be linked by constraints via *state variables* - variables used to describe the mathematical state of a dynamical system. Naturally, as a problem's dimensionality expands, so does the number of required state variables. Unfortunately, SDDP's efficiency diminishes when tasked with handling an extensive number of state variables and begin to face significant computational and convergence challenges. For instance, the original flagship application on the Brazilian hydro-thermal power production scheduling problem only has 4 state variables per stage corresponding to the stored energy [water level] of each reservoir. This characteristic, in turn, has raised concerns about its performance in vast state spaces, leading to its subdued and limited utilization in other real-world problems.

The limitations in handling vast or high-dimensional state spaces can be a considerable deterrent for other problems. High-dimensional problems are increasingly common in numerous fields and a method that struggles with such scale naturally finds limited applicability. Consequently, researchers might be inclined to explore alternative methods that better cater to extensive state space requirements.

Moreover, this state space limitation can sometimes lead to suboptimal solutions or expensive computations, which are not always feasible in real-world applications that demand both accuracy and efficiency. As a result, there might be a preference towards methods that offer a more balanced performance across various problem scales. While SDDP undeniably holds benefits and has proven valuable in some problems, the concerns about its performance in extensive state spaces have played a pivotal role in its adoption in other application areas.

5.3 Stage-wise independence limitation

Another fundamental assumption of SDDP is that the random data has to be *stage-wise independent* - new information becoming available at the time of decision does not depend on the history of the random process. This assumption, while simplifying the computational complexity in certain cases, becomes conflicting in the context of real-world applications. Many systems and processes, especially those of an economic, financial, or natural nature, inherently possess temporal dependencies, making them Markovian or semi-Markovian structure.

Additionally, the stage-wise independence assumption can limit the accuracy and relevance of SDDP when applied to problems with evident temporal dependencies, potentially resulting in sub-optimal solutions. Thus, there's a risk of overlooking critical dependence, which can lead to significant discrepancy in decisions derived from the model. This limitation can avert researchers from utilizing SDDP in problems where capturing the dynamics of random data over time is crucial.

Nonetheless, to address this issue, the work by Löhndorf and Shapiro (2019) introduced two approaches to include dependence of the random process - (1) to model the data process as an auto-regressive process and to add the time series transition equations and (2) another approach based on Markov Chain discretization of the Markovian data process using optimal quantization. Nonetheless, the first approach is restricted only to right-hand side uncertainty and linear cases, and the second approach, although allows modeling any parameter as Markovian data process, has no convergence guarantees.

Hence, while SDDP holds significant potential in many optimization problems, the stage-wise independence assumption tempered on its universal applicability, steering its use toward more restricted domains.

5.4 Data availability

When a parameter of a on optimization problem is random or uncertain, trying to understand the distribution of this parameter can be a daunting challenge. This complexity is further highlighted especially if data is not readily available. The availability, or lack thereof, of data not only impedes precise modeling but also limits the accuracy and reliability of any derived solution. This phenomenon is universally observed across any problems in the field of decision-making under uncertainty for real-world applications. MSPs naturally falls into this category, where the success of the model often depend on the data for uncertain parameters. Consequently, SDDP methods, which operate under these conditions, might see restricted application in practice due to the inherent challenges posed by data limitations.

5.5 Extensions and outlook

Provided the challenges and potential obstacles presented above, we remain optimistic about SDDP's potential to overcome current limitations and address more complex problems. While its major applications have been in the hydrothermal energy sector, finance, and operations management, here are some other domains and specific applications where SDDP can be potentially applied:

Healthcare: The healthcare domain is rife with many complex optimization problems. This includes hospital resource allocation, optimizing treatment plans for chronic illnesses over multiple periods, and surgery scheduling. The uncertainties associated with disease spread, patient outcomes, and resource constraints make SDDP a potential tool for driving productivity and efficiencies.

Disaster Preparedness and Response: Natural disasters, e.g., thunderstorms, earthquakes, and volcanic eruptions, SDDP can be effective in optimizing resource allocation for disaster preparedness and formulating real-time response policies as events unfold.

Environmental Conservation: Optimization problems that involve habitat restoration, wildlife protection, or pollution control over multiple periods, SDDP can help devise policies that account for the uncertainties in ecological settings.

Urban Planning: City infrastructures are complex and interconnected. Whether it is deciding on networks for public transportation, policies for waste management, or land-use planning over different time horizons, SDDP has the potential to handle these problems.

Telecommunications: As communication needs and technologies rapidly evolve, there is a need to optimize network designs, resource allocations, and service over time. SDDP can be a useful tool in making these decisions.

Education: Some common university and educational institution problems are course scheduling, resource allocation (e.g., classrooms or laboratory facilities), and even strategic decisions like opening new programs or campuses.

Marine and Fisheries Management: SDDP can be beneficial from optimizing fishing schedules to route planning for marine vessels to avoid ecologically sensitive areas.

6 Concluding Remarks

Since its inception in 1991, SDDP has, for over three decades, demonstrated a phenomenal performance and is still regarded as one of the go-to methods to solve MSPs. This survey presents a review of the practical applications of SDDP in the literature. A total of 186 significant publications, from 1991 to 2022, were examined from an in-depth search in multiple databases. Various advantages and challenges associated with SDDP were examined to pave way for an in-depth exploration of the literature, in the lens of these different applications.

While its applications span to various fields, the energy sector has notably emerged as its predominant user, with approximately 76% of the articles reviewed deploying the SDDP methodology. This is followed by finance, operations management, and water resources domains. Delving deeper, the hydro-thermal power system production surfaced as the predominant domain, constituting roughly 60% of the literature. This is trailed behind by renewable energies, energy market, trading, and investments, energy storage devices, and grid management applications. This analysis is further enriched with an insightful discussion on the factors influencing the adoption of SDDP across distinct application areas. We have identified four factors - historical background, state space limitation, stage-wise independence assumption limitation, and data availability. Finally, given the ongoing research aimed at augmenting, developing, and enhancing SDDP, we remain optimistic about its potential to overcome current limitations and address more complex problems faced by the state-of-the-art.

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A Detailed reference classification

Table 2: Reference classification (Hydro-thermal power production) - Brazil and Norway

Location	Operation	Reference
Brazil	Long-term	Pereira and Pinto (1991), Gorenstin et al. (1992), Mello et al. (1997), da Silva and Finardi (2003), Maceira and Damázio (2006), Homem-de Mello et al. (2011), Zambelli et al. (2011), Guigues and Sagastizábal (2012), de Matos and Finardi (2012), Shapiro et al. (2013a), Guigues and Sagastizabal (2013), Pinto et al. (2013), Philpott et al. (2013), Shapiro et al. (2013b), Guigues (2014), Calili et al. (2014), Maceira et al. (2014), Oliveira et al. (2015), Brandi et al. (2015), de Castro et al. (2015), De Matos et al. (2015), Ferreira et al. (2015), Lohmann et al. (2016), Brigatto et al. (2017), Rego et al. (2017), Soares et al. (2017), De Matos et al. (2017), Street et al. (2017), de Faria and Jaramillo (2017), Brandi et al. (2017), Löhndorf and Shapiro (2019), Van Ackooij et al. (2019), Shapiro and Ding (2020), Machado and Bhagwat (2020), Treistman et al. (2020), Diniz et al. (2020), Liu and Shapiro (2020), Street et al. (2020), Street et al. (2020), Beltrán et al. (2020), Fredo et al. (2021), Machado et al. (2021), Shapiro and Cheng (2021), Resende et al. (2021), Dowson et al. (2022a), Larroyd et al. (2022), Ávila et al. (2022)
	Medium-term	Löhndorf et al. (2013),Maceira et al. (2014), de Matos et al. (2015), Fredo et al. (2019), da Silva Fernandes et al. (2019), Treistman et al. (2020), Machado et al. (2021), Beltrán et al. (2021), Guigues et al. (2021), Siddig and Song (2022),Borges et al. (2022), Dornellas et al. (2022), Pedrini and Finardi (2022)
	Short-term	Beltrán et al. (2021), Borges et al. (2022)
Norway	Long-term	Rotting and Gjelsvik (1992)
	Medium-term	Mo et al. (2001), Kristiansen (2006), Helseth and Braaten (2015), Helseth et al. (2016), Hjelmeland et al. (2018), Hjelmeland et al. (2019), Scheben et al. (2020), Helseth and Mo (2022), Helseth et al. (2022)
	Short-term	- - -

B Detailed reference classification

Table 3: Reference classification (Hydro-thermal power production) - Other locations

Location	Operation	Reference
New Zealand	Long-term	Duque and Morton (2020)
	Medium-term	Philpott and De Matos (2012), Philpott et al. (2018), Downward et al. (2020), Dowson and Kapelevich (2021)
	Short-term	Pritchard (2015)
Central America	Long-term	Flach et al. (2010), Marques and Tilmant (2013), Rebennack (2014)
	Medium-term	Rebennack et al. (2011), Rebennack (2016), Steeger and Rebennack (2017)
South America	Long-term	Raby et al. (2009), Sauma et al. (2011), Toledo et al. (2015), Morillo et al. (2020), Morillo et al. (2022)
Canada	Long-term	Pina et al. (2017), Mbeutcha et al. (2021)
	Medium-term	Zhang and Ponnambalam (2005), Guan et al. (2018), Côté and Arsenault (2019)
Nile River	Long-term	Goor et al. (2011), Kahsay et al. (2019), Tariku et al. (2021)
Iberian Peninsula	Long-term	Pereira-Cardenal et al. (2016)
	Medium-term	Macian-Sorribes et al. (2017), Leclère et al. (2020)
Senegal	Long-term	Espanmanesh and Tilmant (2022)
	Medium-term	Raso et al. (2017), Raso et al. (2020)
Turkey/Iran	Long-term	Tilmant et al. (2007), Tilmant and Kelman (2007)
	Medium-term	Poorsepahy-Samian et al. (2016)
China	Short-term	Li et al. (2022)
Others	Medium-term	Borges (2022), Cerisola et al. (2012)
	Long-/Short-term	Alvarez et al. (2017), Alvarez et al. (2018)

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